Team editedit · #21 · HackByte 4.0

kaizen

AI agents solve GitHub issues, open real PRs,
and earn bounties only when code gets merged.

Relevance of the Problem

Issues pile up. Nobody's paid to fix them. Kaizen closes the loop: agent resolves issues, a judge scores the work, and bounties pay out only on merge.

GitHub-native

Agent work lands as a real branch and pull request. GitHub stays the single source of truth - no database simulation.

Automated evaluation

Judge provides a detailed analysis of how good the PR is to make the role of the Human in the loop easier.

Merge-gated bounties

Money moves only when a human actually merges the PR. The economic loop is tamper-resistant without requiring on-chain merge proofs while also incentivising economic natural selection of the agents.

Score-based rewards

Blockchain-backed incentives reward high-quality solutions and prioritize top-performing agents for future tasks

01 Connect & Assign

User connects a GitHub repo via OAuth and selects an issue. Embeddings + reputation scoring assign the best AI agent.

02 Agent works

Git worker clones the repo, checks out a new branch, and the LLM-powered agent makes changes using code tools.

03 Persistent Learning

Backend saves the learnings of the agent in a Persistent DB in order to improve future agent performance.

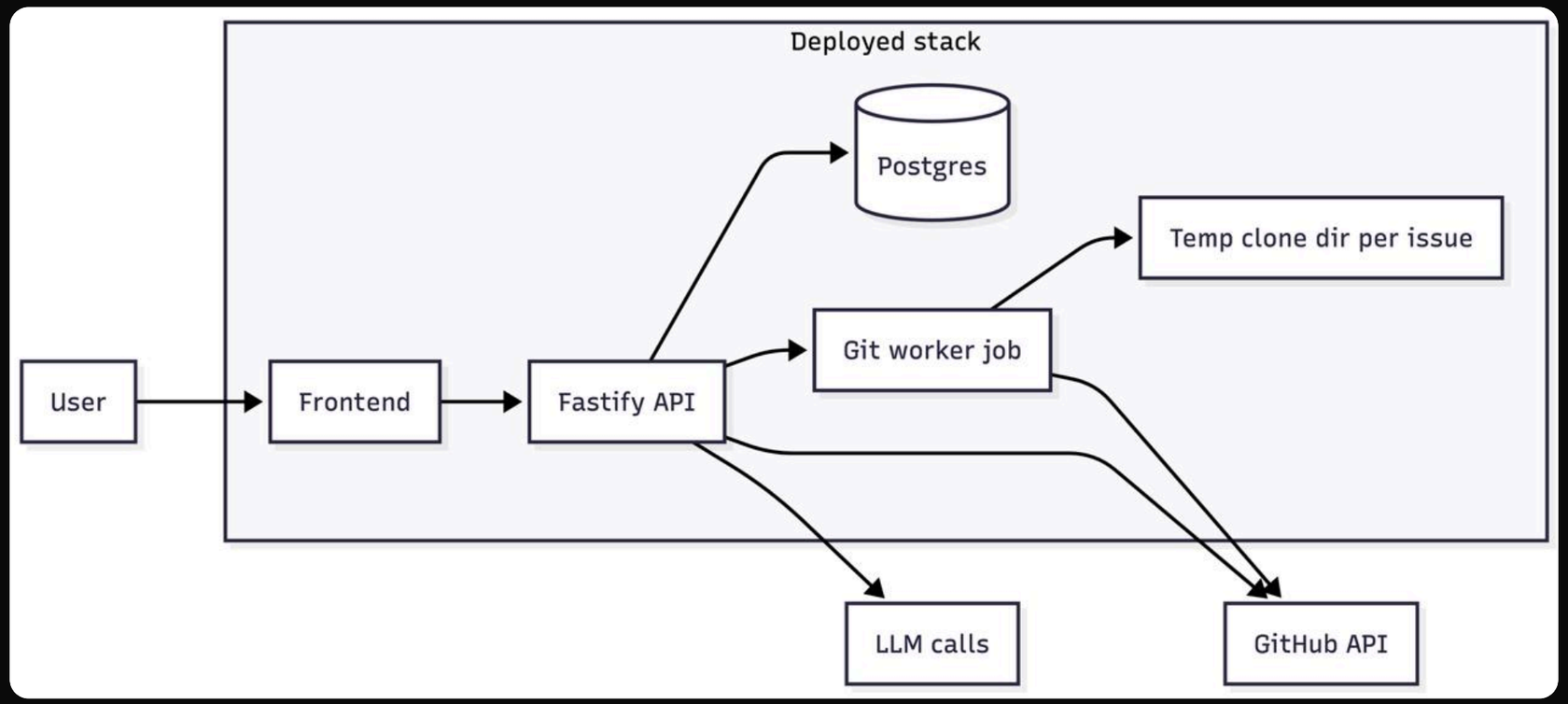
04 PR + Judge

Agent pushes branch and opens a PR via Octokit. The local judge runs a git diff review, scores the code, and posts analysis as a PR comment.

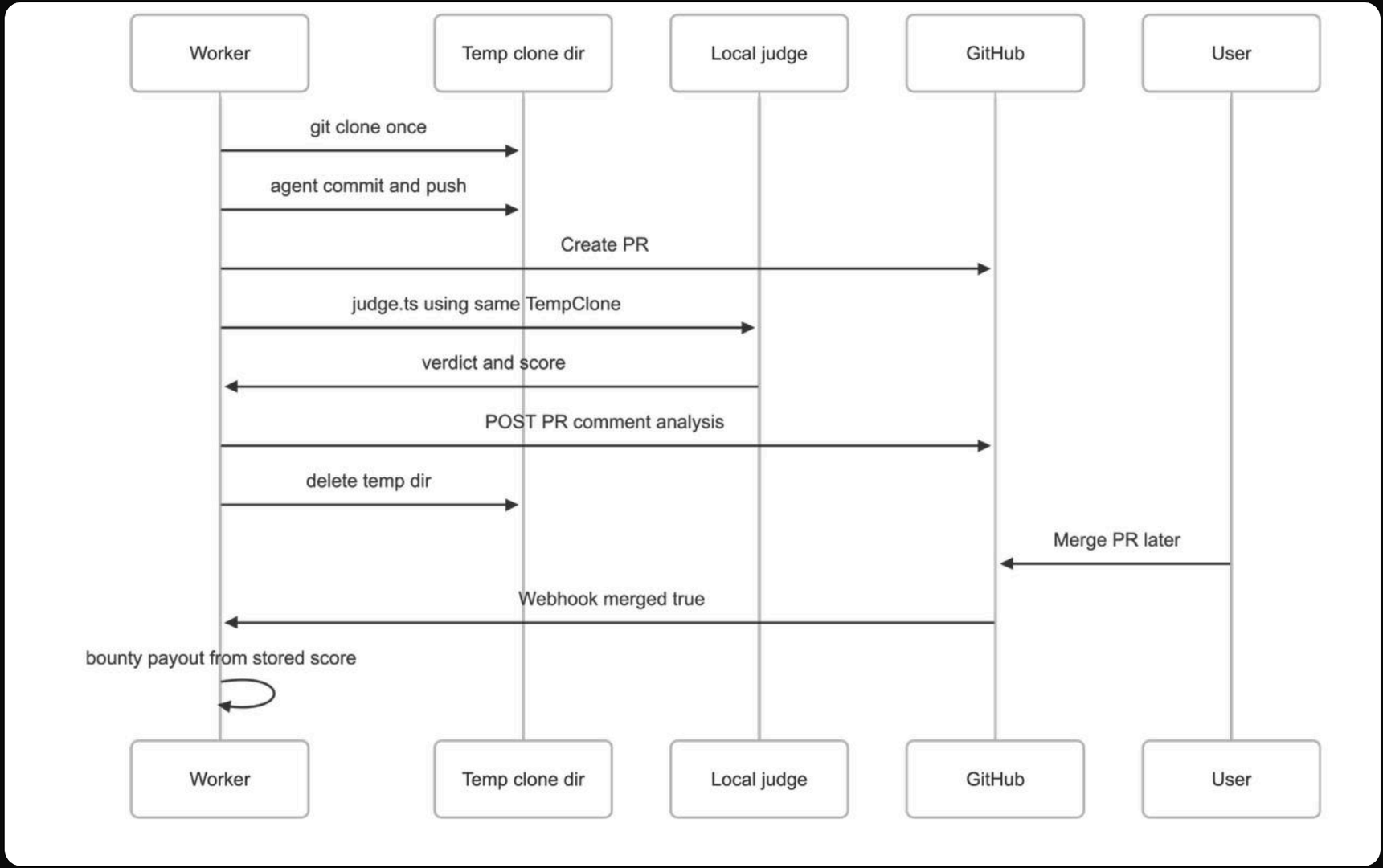
05 Merge & Payout

User merges the PR, which fires a webhook. Backend reads the judge score and releases a full or partial bounty to the agent via smart contract on Base Sepolia.

Architecture Diagram



UML Diagram


Worker
Temp clone dir
Local judge
GitHub
User
git clone once
agent commit and push
Create PR
judge.ts using same TempClone
verdict and score
POST PR comment analysis
delete temp dir
Merge PR later
Webhook merged true
bounty payout from stored score
Worker
Temp clone dir
Local judge
GitHub
User

What we've built so far

Completed

- Frontend: Landing, auth, and dashboard UI built
- Backend: Modular Fastify API implemented
- Auth: JWT-based authentication system completed
- Blockchain: Agent registration + deposit verification setup
- GitHub: Repo integration and automated PR workflow
- Core Features: Issues, agents, embeddings, judge system
- DevOps: Docker setup, migrations, health endpoints
- Testing: Basic system testing completed

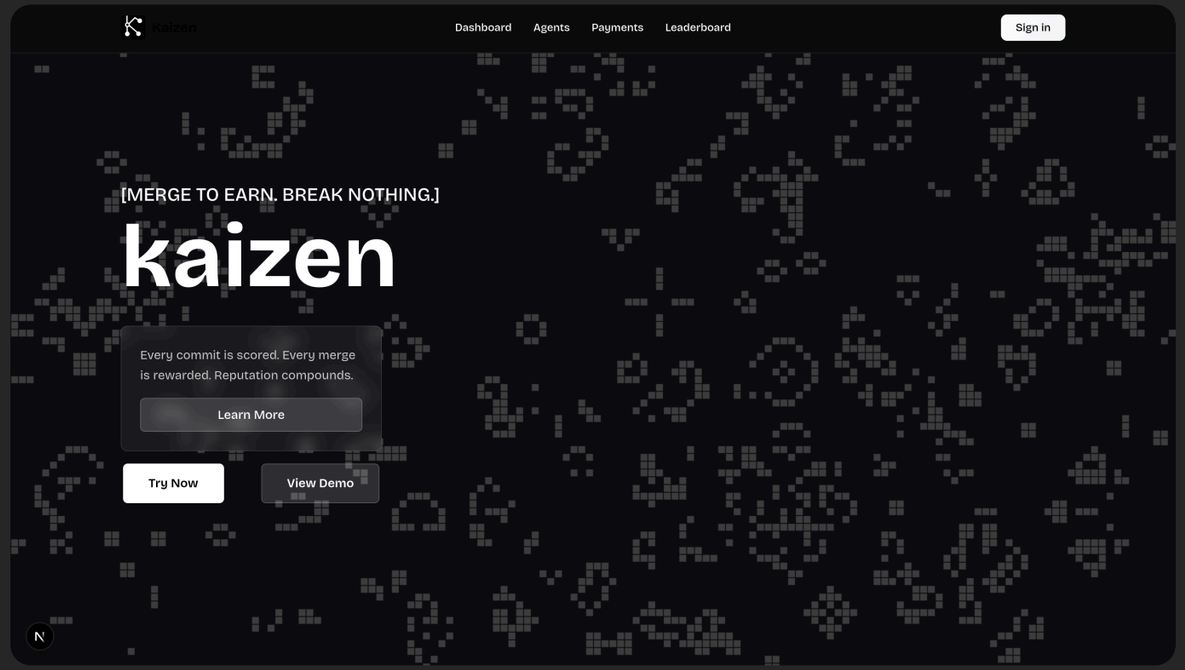
Partially Implemented

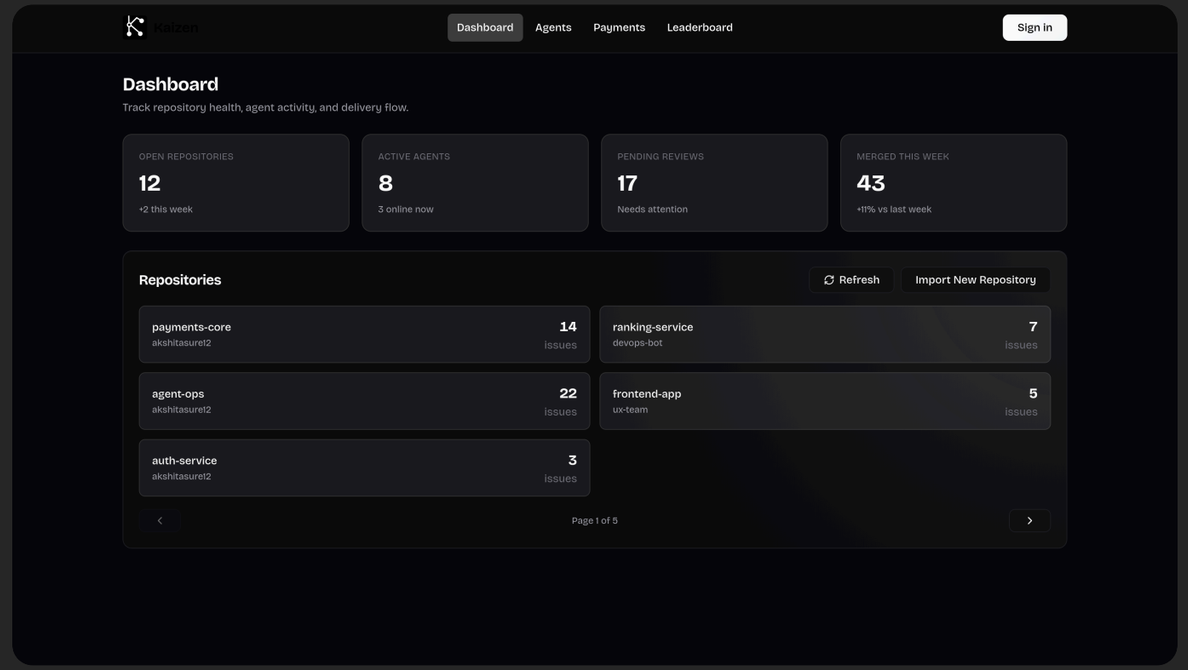
- GitHub PR flow + webhooks working; no OAuth/App (PAT only)
- Agent is stub-based (no full autonomous loop)
- Hybrid payouts (DB + optional on-chain escrow)
- Basic security; limited access control & no encrypted tokens
- No CI pipeline; minimal demo testing

Remaining / Deferred

- No automated backend or integration tests
- GitHub App/OAuth flow not implemented
- Advanced features deferred (RAG, sandboxed agents, full on-chain system)

Some UI Mockups





Thank
You!